Recap on DFT

Signals and Linear Systems

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Outline

- **Review:** CTFS, DTFS, and the DFT
- **Aperiodic Signals:** The DTFT and its relationship to the DFT
- **Zero-Padding:** Increasing frequency-domain resolution
- **The Need for Speed:** Naive DFT complexity
- **Fast Fourier Transform (FFT):**
  - Radix-2 Decimation-in-Time (DIT)
  - Radix-2 Decimation-in-Frequency (DIF)
- **Conclusion:** FFT Complexity Analysis

- Let's start with a Continuous-time periodic signal $x(t)$ with period T_0:

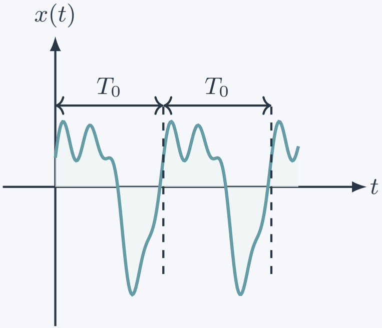
$$x(t+T_0) = x(t), \quad \omega_0 \triangleq \frac{2\pi}{T_0}.$$

- We have learnt the Continuous Time Fourier Series (CTFS) representation is:

CTFS Pair

$$x(t) = \sum c_k e^{jk\omega_0 t}$$

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$



We are now interested in a representation for discrete signals. We can sample $x(t)$ to get a discrete signal $x[n]$.

- Sample $x(t)$ uniformly with sampling period T_s.
- Assume T_0 is captured by N samples:

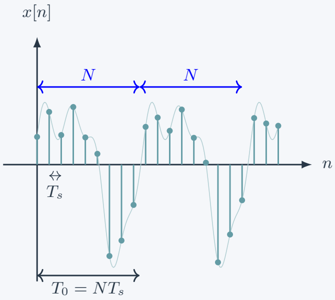
$$T_0 = NT_s \quad (N \in \mathbb{Z}^+).$$

- The purely discrete sequence $x[n]$ is defined as:

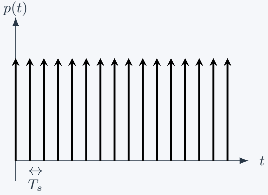
$$x[n] \triangleq x(nT_s), \quad n \in \mathbb{Z}.$$

- Thus $x[n]$ is inherently N**-periodic**:

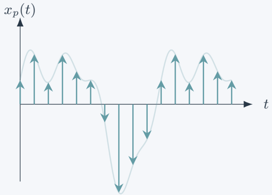
$$x[n+N] = x[n].$$



- We can represent the sampled signal mathematically using an impulse train (aka Dirac comb).
- Multiplying the continuous signal $x(t)$ with the impulse train $p(t)$ gives us the sampled signal $x_p(t)$.
- Each sample $x[n]$ is weighted by a Dirac delta function at time nT_s.



Impulse train $p(t) = \sum \delta(t - nT_s)$.



Sampled signal $x_p(t) = x(t)p(t)$.

$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x[n]\,\delta(t - nT_s).$$

CTFS of the Sampled Signal

Now let us find the CTFS of the sampled signal $x_p(t)$.

- Let c_k denote the CTFS coefficients of the T_0-periodic sampled signal $x_p(t)$:

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, \qquad c_k = \frac{1}{T_0} \int_{T_0} x_p(t)\, e^{-jk\omega_0 t}\, dt.$$

- Substitute $x_p(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s)$ into the analysis equation:

$$\begin{aligned} c_k &= \frac{1}{T_0} \int_{t_0}^{t_0+T_0} \left(\sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s) \right) e^{-jk\omega_0 t}\, dt \\ &= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] \int_{t_0}^{t_0+T_0} \delta(t - nT_s) e^{-jk\omega_0 t}\, dt \\ &= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n]\, e^{-jk\omega_0 n T_s} \\ &= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}. \end{aligned}$$

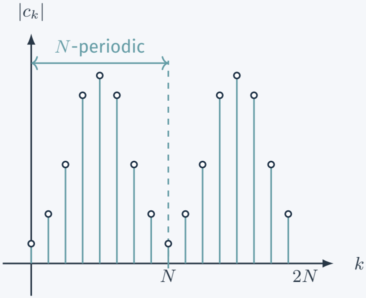
We can take $t_0 = -\frac{T_s}{2}$, so $\delta(t - nT_s)$ is non-zero in $[t_0, t_0 + T_0]$ only for $n = 0, 1, 2, \ldots, N-1$.

Periodicity of c_k

- Examine the exponent in c_k:

$$c_{k+N} = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}(k+N)n}$$
$$= c_k \underbrace{e^{-j2\pi n}}_{=1} = c_k$$

- The contiguous frequency coefficients are N-periodic!
- **Only N unique frequency values** exist before the spectrum repeats.
- Leads to the N-point Discrete Fourier Transform.



- The N-point **discrete Fourier transform (DFT)** of a length-N sequence $\{x[n]\}_{n=0}^{N-1}$ is defined simply as a scaled version of one period of the unique c_k components:

Discrete Fourier Transform (DFT)

$$X[k] \triangleq \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}, \qquad k = 0, 1, \ldots, N-1$$

- Thus, the relationship is $c_k = \frac{1}{T_0} X[k]$.

- We cannot simply use the CTFS synthesis equation to recover our signal:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

- This equation evaluates to the continuous-time impulse train $x_p(t)$, not the purely discrete sequence $x[n]$ which we desire.

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- However, there is a general pattern in transforms from our experience:
  - **Forward Transform** (Analysis) usually uses $e^{-j\cdots}$ (e.g., $e^{-jk\omega_0 t}$)
  - **Inverse Transform** (Synthesis) usually uses $e^{+j\cdots}$ (e.g., $e^{+jk\omega_0 t}$)
- Let's apply this intuitive pattern to derive an explicit recovery formula for $x[n]$ using $e^{+j\cdots}$.

Deriving the Inverse DFT (1/2)

- Multiply both sides of the DFT by $e^{j\frac{2\pi}{N}kn}$ and sum over k:

$$\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn} = \sum_{k=0}^{N-1}\left(\sum_{m=0}^{N-1} x[m]e^{-j\frac{2\pi}{N}km}\right)e^{j\frac{2\pi}{N}kn}$$
$$= \sum_{m=0}^{N-1} x[m]\left(\sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}k(n-m)}\right)$$

- Evaluate the inner summation using the geometric series sum formula. For $n \neq m$:

$$\sum_{k=0}^{N-1}\left(e^{j\frac{2\pi}{N}(n-m)}\right)^k = \frac{1-e^{j2\pi(n-m)}}{1-e^{j\frac{2\pi}{N}(n-m)}} = \frac{1-1}{1-e^{j\frac{2\pi}{N}(n-m)}} = 0$$

Deriving the Inverse DFT (2/2)

- And for $n = m$, the summation simply evaluates to:

$$\sum_{k=0}^{N-1} e^{0} = N \implies \sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}k(n-m)} = \begin{cases} N & n = m \\ 0 & n \neq m \end{cases}$$

- Applying this to the previous equation, all terms in the outer sum become zero except when $m = n$:

$$\sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn} = x[n] \cdot N$$

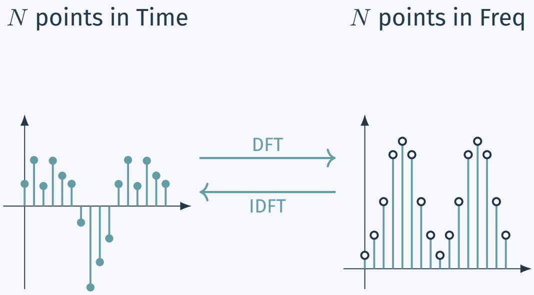
- Rearranging for $x[n]$ gives the **Inverse DFT (IDFT)**:

Inverse Discrete Fourier Transform (IDFT)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j\frac{2\pi}{N}kn}, \qquad n = 0, 1, \ldots, N-1$$

Recap: The DFT for Periodic Signals

- We started with a continuous-time periodic signal $x(t)$.
- Sampling it yielded a **periodic discrete sequence** $x[n]$ with period N.
- Its frequency representation c_k is uniquely defined by exactly N contiguous components.
- Ultimately, the **DFT** transforms exactly N points of $x[n]$ into N unique frequency coefficients $X[k]$.
- The **IDFT** perfectly reconstructs those N points in time from these N frequency values.

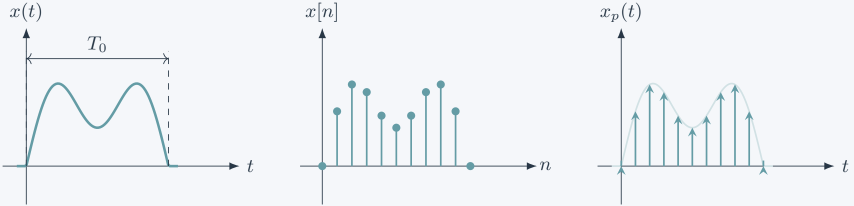


What about Aperiodic Signals?

But does DFT only apply to periodic signals? What if the signal is aperiodic?

- Let's consider an aperiodic signal $x(t)$ restricted to a finite interval T_0.
- Uniformly sampling this yields a finite-length sequence $x[n]$ that is nonzero only for $0 \leq n \leq N-1$.
- Using the pulse train sampling method, we can represent the sampled signal as:

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT) = \sum_{n=0}^{N-1} x[n]\delta(t - nT)$$



From the Sampled Signal to the DTFT

- First take the CTFT of the sampled impulse train:

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT}$$

- Now compress the frequency axis by the sampling period, i.e. apply the mapping $\omega T \mapsto \omega$.
- Under that relabeling, the same spectrum becomes the **Discrete-Time Fourier Transform (DTFT)**:

Discrete-Time Fourier Transform (DTFT)

$$X(e^{j\omega}) \triangleq X_p(j\omega)\big|_{\omega T \mapsto \omega} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

- Hence, the DTFT is the Fourier transform of the sampled signal, with a compressed frequency axis.
- For our length-N capture,

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}.$$

Periodicity of the DTFT

Period of the DTFT

The value of $X(e^{j\omega})$ repeats every 2π:

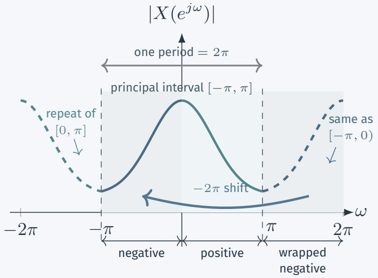
$$e^{-j(\omega+2\pi)n} = e^{-j\omega n}e^{-j2\pi n}$$
$$= e^{-j\omega n}$$

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

How to Read One Period

- Any interval of width 2π contains one complete DTFT period.
- A common principal interval is $[-\pi, \pi]$.
- $[\pi, 2\pi)$ is the shifted copy of $[-\pi, 0)$.

Before $\omega T \mapsto \omega$, the physical sampled spectrum has period $\omega_s = \frac{2\pi}{T}$; we revisit that view in Sampling.



The Path to the DFT (Aperiodic Signals)

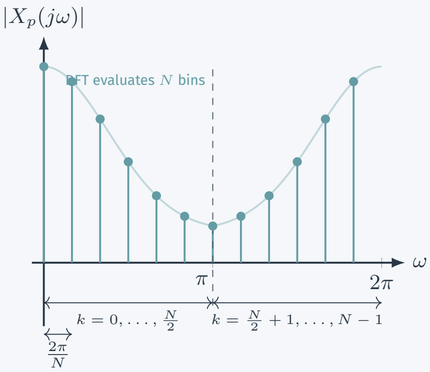
- Recall the DFT definition:

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\,\frac{2\pi}{N}\,kn}.$$

- We see that the DFT evaluates $X(e^{j\omega})$, the DTFT, at **discrete frequencies** $\omega = \frac{2\pi}{N}k$:

$$X[k] = X(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}.$$

- So the N-point DFT takes N **evenly spaced samples** over one repeated spectrum.
- For even N, bins $k = 0, \ldots, \frac{N}{2}$ run from dc to the highest positive frequency, while bins $k = \frac{N}{2}+1, \ldots, N-1$ run from large negative frequencies back toward dc after wrapping.



Zero-Padding for Increased Frequency Resolution

- **Zero-Padding**: For an aperiodic $x[n]$ of length N, we can artificially append zeros to define it over a total length $M \geq N$.
- Taking the M-point DFT samples the same continuous DTFT spectrum $X(e^{j\omega})$, but at a much finer resolution of $2\pi/M$ over that same wrapped 2π frequency axis.